

EduGenie – Development Overview

Project: EduGenie – Google Gemini Powered Learning Assistant

1. Introduction

EduGenie is a web-based AI learning assistant designed to help students understand educational topics, practice through quizzes, summarize study material, and receive learning recommendations.

The application uses a **FastAPI backend** with an **HTML and CSS frontend** and connects to the **Google Gemini API** for AI-powered processing.

2. Technologies Used

- **Programming Language:** Python
- **Backend Framework:** FastAPI
- **Web Server:** Uvicorn
- **Frontend:** HTML and CSS
- **AI Service:** Google Gemini API
- **Template Engine:** Jinja2
- **Version Control:** Git and GitHub

3. Main Development Components

3.1 Backend

The FastAPI backend receives requests from the web interface and processes them using the appropriate learning functionality.

3.2 Learning Features

EduGenie provides the following learning functions:

- Question and Answer
- Simplified Explanation
- Quiz Generation
- Text Summarization
- Learning Path Recommendation

3.3 Frontend

The frontend provides a simple interface where students can select a learning task, enter their input, submit the request, and view the generated result.

3.4 AI Integration

The Google Gemini API is used to generate AI-powered responses for the learning tasks supported by the application.

4. Development Workflow

The development process follows:

Frontend → FastAPI Backend → Learning Module → Google Gemini API → Backend → Frontend

5. Project Structure

The main project components include :

```
EduGenie/
|
|— main.py
|— qna.py
|— explanation_module.py
|— quiz_module.py
|— summary_module.py
|— learning_path.py
|
|— templates/
|   |— index.html
|
|— static/
|   |— style.css
```

6. Development Outcome

The development phase produces a working AI-powered learning assistant that provides students with multiple learning support functions through a web interface.

7. Conclusion

The EduGenie development combines a FastAPI backend, web frontend, separate learning modules, and Google Gemini AI services to create an accessible learning assistant for students.